

Human-AI Collaborative Web Engineering: Orchestration, Hallucination Auditing, and Domain Governance

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LIVE DEPLOYMENT

d1500707-portfolio.netlify.app

1. Objective & AI Delegation: In this assignment, my objective was to partner with generative AI to build a responsive, multi-section portfolio articulating my dual identity as a doctoral clinical AI researcher and an enterprise technology executive. I tasked the AI with translating information architecture requirements into clean, semantic HTML and responsive Tailwind CSS utilities, scaffolding mobile navigation drawers, and generating comparative visual components without relying on fragile external libraries.

2. AI Strengths & Accelerated Execution: Generative AI functioned as an exceptional cognitive catalyst and rapid prototyper. It constructed cohesive typographic hierarchies, calculated grid layouts, and drafted flexbox structures in seconds. By offloading front-end boilerplate coding to AI, development velocity increased fourfold, freeing my cognitive bandwidth to concentrate on narrative coherence, visual proof selection, and rigorous technical positioning.

3. Critical Vulnerabilities & AI Hallucinations: Left unguided, AI demonstrated profound deficiencies in factual accuracy, contextual prudence, and safety alignment. Factually, the model hallucinated an unearned master's degree (NCCU Digital Content) and fabricated an irrelevant corporate lecture for a non-existent parking association. Technically, it generated incorrect file extensions (.jpg instead of .png), introducing potential 404 image errors. Most alarmingly, in medical framing, AI characterized my clinical rehabilitation model as an "autonomous diagnostic agent," violating fundamental clinical bioethics and regulatory safety norms.

4. Human Judgment & Critical Interventions: Human oversight proved essential in transforming plausible output into verified truth. I conducted a rigorous biographical audit, restoring my engineering lineage from Tatung University (B.S. and M.S. in Materials Engineering) and authentic keynote engagements with trade bodies like the Taipei Office Automation Equipment Association, Food & Pharma Machinery Association, and TAITRA. Clinically, I re-anchored the architecture to a physician-supervised, human-in-the-loop decision support system bounded by Causal Forest inference. Crucially, I infused lived authenticity: articulating the tragedy of family sarcopenia and integrating my empirical N=1 AI-assisted body recomposition data (86.5 kg to 70.4 kg; blood pressure normalization) as tangible clinical motivation.

5. Key Learnings on Human-AI Collaboration: This project reinforced that generative AI is a high-speed synthesizer, not an authoritative intellect. It generates plausible surface structure but lacks epistemic accountability and ethical discernment. True engineering mastery lies in the human's capacity to serve as an unyielding domain gatekeeper—challenging assumptions, correcting hallucinations, and embedding genuine lived experience. In the generative era, critical human judgment remains the ultimate determinant of rigor, credibility, and safety.

CORE SYNTHESIS & HUMAN JUDGMENT TAKEAWAY

Generative AI accelerates visual and syntactical scaffolding; however, domain accuracy, clinical safety boundaries, and biographical truth remain non-delegable human responsibilities. Critical oversight converted algorithmic hallucinations into a verifiable, high-impact scholarly portfolio.